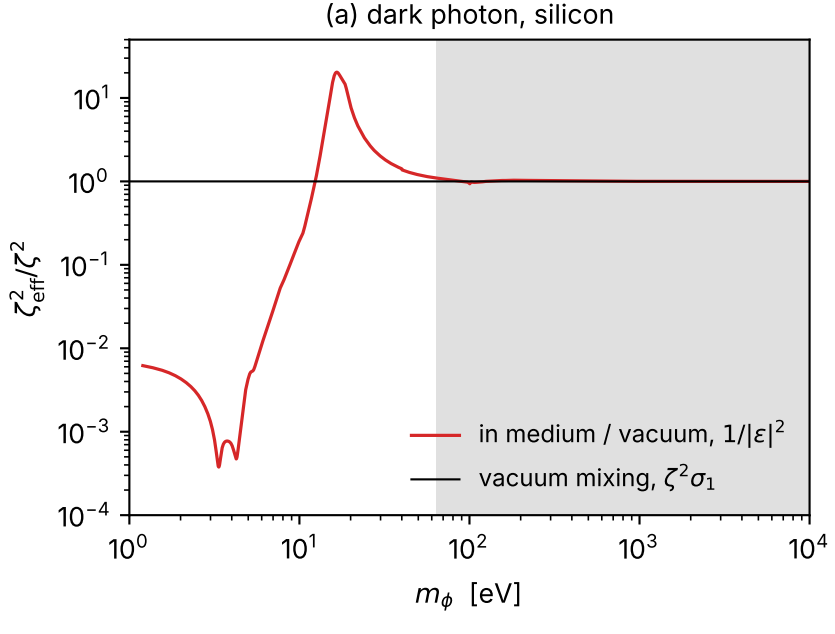
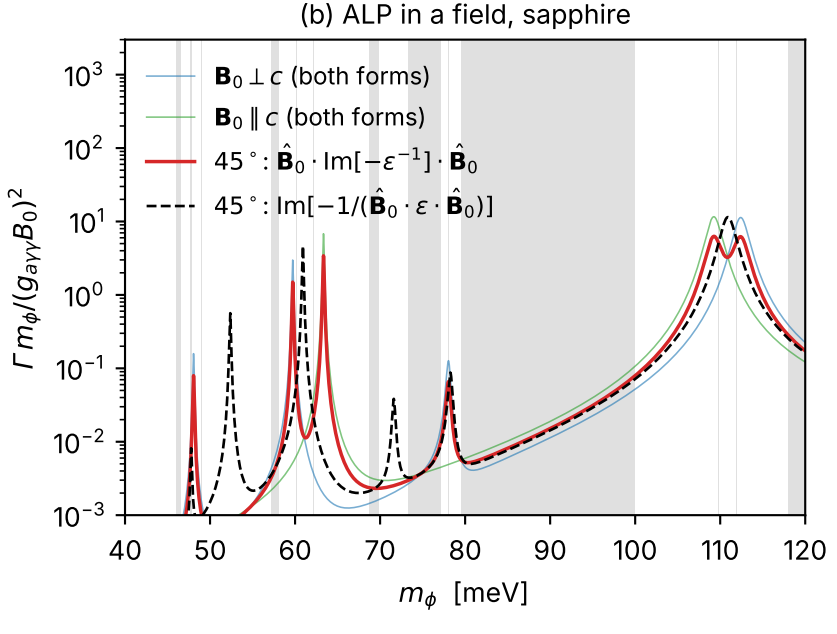

(a) dark photon, silicon
$\zeta^2_{\rm eff}/\zeta^2$
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
in medium / vacuum, $1/|\varepsilon|^2$
vacuum mixing, $\zeta^2\sigma_1$
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
m_ϕ [eV]


(b) ALP in a field, sapphire
$\Gamma\, m_\phi/(g_{a\gamma\gamma}B_0)^2$
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
$\mathbf{B}_0 \perp c$ (both forms)
$\mathbf{B}_0 \parallel c$ (both forms)
$45^\circ: \hat{\mathbf{B}}_0 \cdot \mathrm{Im}[-\varepsilon^{-1}] \cdot \hat{\mathbf{B}}_0$
$45^\circ: \mathrm{Im}[-1/(\hat{\mathbf{B}}_0 \cdot \varepsilon \cdot \hat{\mathbf{B}}_0)]$
40
50
60
70
80
90
100
110
120
m_ϕ [meV]